

8.4 Kites and Parallelograms

Lesson Introduction

 Benchmark	MA.912.GR.1.4 Prove relationships and theorems about parallelograms. Solve mathematical and real-world problems involving postulates, relationships, and theorems of parallelograms.		 Learning Target	I can use properties of parallelograms, kites, and special parallelograms to solve mathematical and real-world problems.
 Benchmark Coherence	Building On		Working Toward	
	N/A		N/A	
 Essential Questions	- What is a kite? - Is a kite a special type of parallelogram?			
 Lesson Narrative	Building on the properties of a parallelogram, in this lesson, students will explore the properties of kites. Students will discover that kites are not a type of parallelogram after classifying descriptions of sides, angles, and diagonals for different quadrilaterals. Using the properties of parallelograms and kites, students will solve mathematical and real-world problems.			
 Component Summary	8.4.1 Warm-Up (~5 min.)	Estimate the number of cans in an artistic structure. (SE p. 26)	8.4.4 Guided Instruction (10-15 min.)	Solve problems using the properties of kites and parallelograms. (SE p. 29)
	8.4.2 Exploration (5-10 min.)	Determine the properties of a kite. (SE p. 26)	8.4.5 Your Turn! (5-10 min.)	Solve problems using the properties of kites and parallelograms. (SE p. 31)
	8.4.3 Collaboration (5-10 min.)	Collaborate to categorize features of different quadrilaterals. (SE p. 28)	8.4.6 Wrap-Up (~5 min.)	Solve a mathematical problem using the properties of a kite. (SE p. 32)
 Resources	Glossary		Materials	
	<u>Key Terms:</u> - Kite <u>Additional Terms:</u> - Quadrilateral - Rectangle - Square - Rhombus		- Ruler - Protractor	
			Additional Preparation	
			Consider making copies of the statements in 8.4.3 Collaboration for students to sort into categories.	

 Teacher Tips	Common Misconceptions • Students may set the wrong segments equal when creating equations. • Students who struggle with fractions may have trouble with 8.4.6 Wrap-Up. Have students show all their steps, so that it can easily be identified if there are issues with the content in the lesson (so you can create a plan for remediation), or if there are calculation errors.	Things to Consider • If time is an issue, then consider assigning 8.4.5 Your Turn! for homework.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder 8.4.1 Warm-Up is an opportunity to inspect a unique figure known as a “canstruction.” Students are encouraged to reflect on their wonderings of the image. Consider extending this to asking what they notice, prior to answering question 2.	MA.K12.MTR.7.1: Real-World Contexts Students are provided a variety of real-world contexts to apply the properties of quadrilaterals they have learning thus far in the unit. In both 8.4.4 Guided Instruction and 8.4.5 Your Turn!, students use the properties of parallelograms, rectangles, and kites to solve mathematical and real-world problems.
 Differentiate Instruction	Consider allowing students the option to complete 8.4.2 Exploration in pairs or individually.	
	Consider modeling for students how to mark the text and diagram while reading the situation in 8.4.4 Guided Instruction, question 3, including thinking aloud about why you’re making those markings. Release to students by reading aloud question 2 in 8.4.5 Your Turn!, while students mark the text and diagram.	
Lesson Notes:		

8.4.1 Warm-Up: Canstruction

Component Overview: Students will read through a context about “Canstructions,” and respond to the prompts of what they wonder and how many cans are used to build the structure.



8.4.1 Warm-Up: Canstruction

Each year just before Thanksgiving there are competitions held all over the world to build unique Canstructions with canned food that is collected from the community. The food is then given to local food banks. The picture shows the Canstruction called “Downside Up,” which was the winning exhibit from Canstruction 2010 in New York City.



1. What do you wonder about “Downside Up”?
Answers will vary. I wonder if there is any reinforcing material, such as glue, holding it together.
2. Make an educated guess at the number of cans used to build “Downside Up”. Show your work.
Answers will vary. I guess that there will be 2,508 cans. There appears to be 4 congruent pyramid structures. The square of the level number is the number of cans on that level in one pyramid. The 1st level has 1 can, the 2nd level has 4 cans, the 3rd has 9, and so on. Add the cans on all 12 levels then multiply that by 4 pyramids to get 2,600 cans. Then, it appears on the top level that each pyramid shares a row with the neighboring pyramid which accounts for sharing 92 cans. Subtracting those gives 2,508 cans.
3. Compare your educated guess with your partner’s. Describe how close your guess was to your partner’s guess.
Answers will vary.



Consider utilizing the Notice and Wonder Routine and have students make note of something they notice in the picture, before moving to question 2.



Students may want to explore if more of these exist. If time permits, consider showing students more “canstruction” creations.

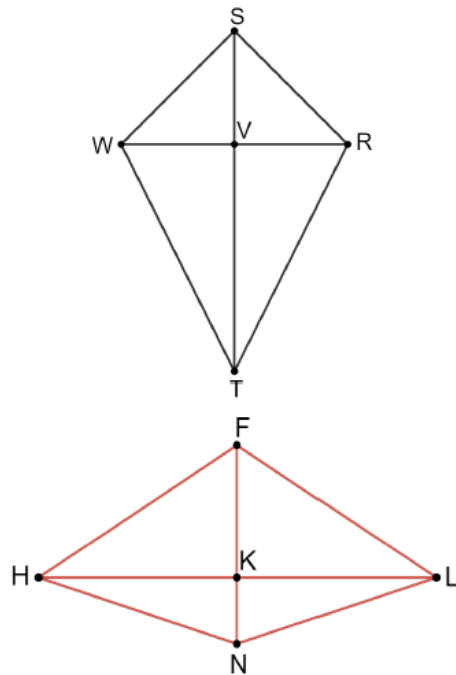
8.4.2 Exploration: What Is a Kite?

Component Overview: Students work collaboratively to measure sides, diagonal lengths, and angles of two kites to determine the properties of this type of quadrilateral.



8.4.2 Exploration: What is a Kite?

Kite $SRTW$ with diagonals $\overline{ST}$ and $\overline{WR}$ intersecting at point V and $FLNH$ with diagonals $\overline{FN}$ and $\overline{HL}$ intersecting at point K are shown.



1. Using a ruler, work with a partner to fill in the chart below with the appropriate measurements. Each partner should find the measurements for one kite and share their results with the other.



This Exploration is scaffolded to allow students to engage with it with their partner, rather than being guided through by the teacher. Students may revise their thinking throughout the Exploration as they engage with each new reflection.

Observe and listen as students work with their partners. Make note of students' ideas so that you can ask them to share with the whole group before moving to the next component.

Additional scaffolding questions are provided, if needed, but this should be a primarily student-directed Exploration.



*In looking at the two kites, are they alike in any way(s)? Explain your thinking.
Do you think either of these shapes fit into the definition of quadrilateral, parallelogram, rectangle, square, or rhombus?*

8.4.2 Exploration: What Is a Kite? (continued)

Measurements depend on student lesson format.

<i>SRTW</i>		<i>FLNH</i>	
Segment	Measurement	Segment	Measurement
$\overline{SR}$	4 cm	$\overline{FL}$	5.1 cm
$\overline{SW}$	4 cm	$\overline{FH}$	5.1 cm
$\overline{RT}$	6.3 cm	$\overline{HN}$	4.45 cm
$\overline{WT}$	6.3 cm	$\overline{LN}$	4.45 cm
$\overline{ST}$	8.4 cm	$\overline{FN}$	4.2 cm
$\overline{SV}$	2.8 cm	$\overline{FK}$	2.8 cm
$\overline{TV}$	5.6 cm	$\overline{NK}$	1.4 cm
$\overline{WR}$	5.6 cm	$\overline{HL}$	8.4 cm
$\overline{WV}$	2.8 cm	$\overline{HK}$	4.2 cm
$\overline{RV}$	2.8 cm	$\overline{LK}$	4.2 cm

- a. What do you notice about $\overline{SR}$ and $\overline{SW}$ in *SRTW* and $\overline{FL}$ and $\overline{FH}$ in *FLNH*?

They are congruent.

- b. What do you notice about $\overline{RT}$ and $\overline{WT}$ in *SRTW* and $\overline{HN}$ and $\overline{LN}$ in *FLNH*?

They are congruent.

2. These segment pairs are **adjacent** sides of a kite.

- a. Make a conjecture about the measures of adjacent sides of a kite.

Two sets of adjacent sides of a kite are congruent.

- b. Make a conjecture about the lengths of diagonals of a kite.

The lengths of the diagonals are not congruent. One diagonal bisects the other (but not vice versa).



While the answer key is in inches, answers in other units, such as centimeters, should be accepted as well.



While monitoring students writing their conjectures, listen for accuracy in their reasoning and wording.

Consider having students mark on the kites from question 1, what they note in their conjectures in question 2.

8.4.2 Exploration: What Is a Kite? (continued)

3. Using a protractor, work with your partner to fill in the chart below with the appropriate measurements. Each partner should find the measurements for one kite and share their results with the other.

<i>SRXW</i>		<i>FLNH</i>	
Angle	Measurement	Angle	Measurement
$\angle SVR$	90°	$\angle FKL$	90°
$\angle WSR$	90°	$\angle HFL$	110°
$\angle RTW$	50°	$\angle LNH$	140°
$\angle TWS$	110°	$\angle FHN$	55°
$\angle TRS$	110°	$\angle FLN$	55°

- a. Make a conjecture about what happens when the diagonals of a kite intersect.

When diagonals intersect, they form a 90° and one diagonal is bisected but the other is not

- b. Make a conjecture about the measures of opposite angles of a kite.

One pair of opposite angles of a kite are congruent.

A **kite** is a quadrilateral with exactly two pairs of adjacent congruent sides.



- A **kite** has the following properties:
- One pair of opposite congruent angles
 - Perpendicular diagonals
 - One bisected diagonal



If making a chart, or if students have a foldable about quadrilaterals, have students add this information about kites.

8.4.3 Collaboration: Quadrilaterals

Component Overview: Students work collaboratively to match statements about quadrilaterals to the type(s) of quadrilateral they represent. Once finished, students will compare their matches with other groups, and decide if a kite can be classified as a parallelogram.



8.4.3 Collaboration: Quadrilaterals

1. With your partner, use the bank of phrases to fill in the chart comparing quadrilaterals. Statements can be used more than once.

A. Opposite sides are congruent.	B. Opposite sides are parallel.
C. All sides are congruent.	D. Diagonals bisect each other.
E. Opposite angles are congruent.	F. Diagonals are congruent.
G. All angles are congruent.	H. Diagonals are perpendicular.
I. Two pairs of adjacent sides are congruent.	J. One diagonal is bisected.
K. One pair of opposite angles are congruent.	L. Opposite angles are supplementary.
M. One pair of parallel sides	N. Non-parallel sides are congruent.
O. Two pairs of congruent angles	

a. Write the letter(s) from the bank.

Quadrilateral	Sides	Angles	Diagonals
Parallelogram	A, B	E, O	D
Rectangle	A, B	E, G, L, O	D, F
Square	A, B, C, N, I	E, G, L, O	D, F, H
Rhombus	A, B, C, N, I	E, O	D, H
Kite	I	K	H, J

b. Compare your chart with that of another pair. Add in anything you may have missed.

Answers will vary.

c. Discuss with your partner whether or not kites can be classified as parallelograms.

Answers will vary.



For students who are struggling with identifying all quadrilaterals for a given statement, consider copying the statements to sorting cards, making duplicates for the statements that repeat. For instance, make four copies of statement A, four copies of statement B, two copies of statement C, etc.



Inform students that all phrases may not be used.



Before moving to the next component, address any questions that arose during the Collaboration.

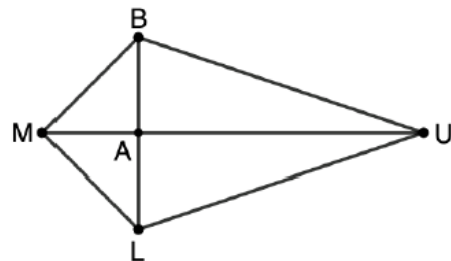
8.4.4 Guided Instruction: Quadrilaterals and Algebra

Component Overview: Students will be guided through using the properties of parallelograms and kites to solve mathematical and real-world problems.



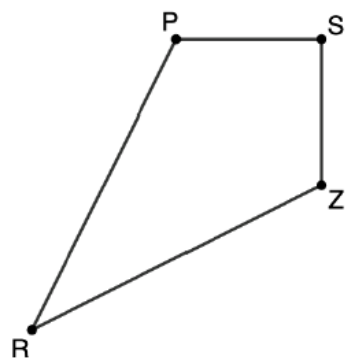
8.4.4 Guided Instruction: Quadrilaterals and Algebra

1. $MBUL$ with diagonals $\overline{BL}$ and $\overline{MU}$ intersecting at point A is shown. Find the value of x that makes $MBUL$ a kite if $BA = 3x - 1$, $LA = 17$, $MA = 2x$, and $UA = 23$.



$$\begin{aligned} BA &= LA \\ 3x - 1 &= 17 \\ x &= 6 \end{aligned}$$

2. Find the values of x and y that make $PSZR$ a kite if $PS = 5x - 4$, $PR = 8y - 4$, $SZ = 3y + 1$, and $RZ = 36$.



$$\begin{aligned} PR &= RZ \\ 8y - 4 &= 36 \\ y &= 5 \end{aligned}$$

$$\begin{aligned} PS &= SZ \\ 5x - 4 &= 3y + 1 \\ 5x - 4 &= 3(5) + 1 \\ x &= 4 \end{aligned}$$

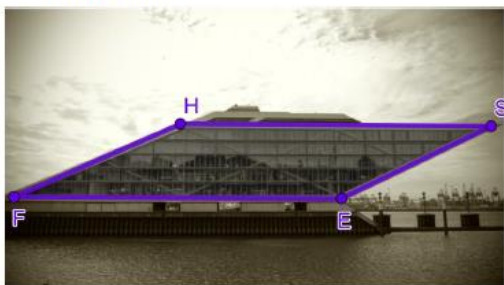


Consider asking the following for question 1, prior to solving for the answer:

- What segments are we given information about?
- What property about kites relate these segments?
- Are both diagonals being bisected?
- Which diagonal is being bisected?
- What does it mean to be bisected?

8.4.4 Guided Instruction: Quadrilaterals and Algebra (continued)

3. The Dockland office building is one of the most famous buildings in Hamburg, Germany. Built in 2005, this building houses roughly 9,000 square meters of office space and breathtaking views of the Elbe River, but the size and the views are not what makes this building so famous. The building itself is in the shape of a parallelogram, $HSEF$.



The perimeter of the Dockland building measures approximately 238 meters. If the length of $\overline{FE}$ is 2.8 times the length of $\overline{FH}$, what are the lengths of $\overline{HS}$, $\overline{SE}$, $\overline{FE}$, and $\overline{FH}$? Round your answers to the nearest thousandth of a meter.

$$FH + SE + HS + FE = 238 \text{ m}$$

$$x + x + 2.8x + 2.8x = 238$$

$$x = 31\frac{6}{19}$$

$$FH \text{ and } SE = 31.316 \text{ m}$$

$$HS \text{ and } FE = 87.684 \text{ m}$$



For question 3, model for students how to mark the text and diagram while reading the situation, including thinking aloud about why you're making those markings. It is important for students to hear the thinking process for this strategy, while also seeing it in action. Marking the text provides students the opportunity to make better sense of the context, while also providing insight for the teacher about what information students are pulling from the text, and what they may be missing.

8.4.5 Your Turn!

Component Overview: Students use the properties of parallelograms to determine if a fence for a given context is a rectangle. Students will also use the properties of kites to solve a real-world problem.



8.4.5 Your Turn!

- Oliver's carpentry company, Wood 4 You, has been hired to build a fence around a vegetable garden. The gate must be a perfect rectangle. The gate Oliver built is shown, where quadrilateral $BYRX$ outlines the gate.



Oliver wants to be absolutely sure that the gate is a rectangle. Describe two different ways Oliver can use his tape measure to ensure the gate is a true rectangle.

Answers will vary.

He could measure to make sure the diagonals were congruent and are bisecting each other.

From one angle, he could make a mark 3 inches (in.) from the vertex on one leg and 4 in. from the same vertex on the other leg of the angle. If he measures the length between these two marks and it is 5 in., then he knows that angle is a right angle. He can then check to make sure the other 3 angles are right angles using the same method.



If time is an issue, then consider assigning Your Turn! for homework.

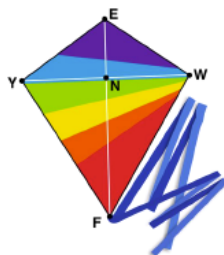


Consider allowing students an alternative to writing a description. Some students may wish to verbally explain to a partner. Have one partner share one of the ways they listed. Their partner can then summarize what they heard, and share a different way to use the tape measure to ensure the gate is a true rectangle.

8.4.5 Your Turn! (continued)

2. Piper is entering a kite-making contest. The kite she designed is shown, where the kite has been labeled as $YEW\!F$ with measurements $EW = 7.5$, $YN = 6$, and $FN = 10.2$, given in inches.

Piper wishes to add a strip of glitter fabric along diagonals $\overline{EF}$ and $\overline{YW}$ that will overlap at their intersection point N , as well as along the perimeter of the kite. How many inches of glitter fabric are needed to cover the diagonals and the perimeter of the kite?



$$\begin{aligned}
 EN &= \sqrt{7.5^2 - 6^2} = 4.5 & EF &= EN + NF = 14.7 \\
 WF &= YF = \sqrt{6^2 + 10.2^2} \approx 11.834 \\
 \text{Amount of fabric} &= YE + EW + WF + YF + YW + EF \\
 \text{Amount of fabric} &\approx 7.5 + 7.5 + 11.834 + 11.834 + 12 + 14.7 \\
 \text{Amount of fabric} &\approx 65.368 \text{ inches}
 \end{aligned}$$



Determine from observational data during the lesson, if a small, teacher-facilitated group is needed to work through question 2.

Another option, consider reading the situation in question 2 aloud while students mark the text and diagram on their own. Then, students could work in pairs to discuss and compare what information they marked, and what they understand about the context, before considering the solution.

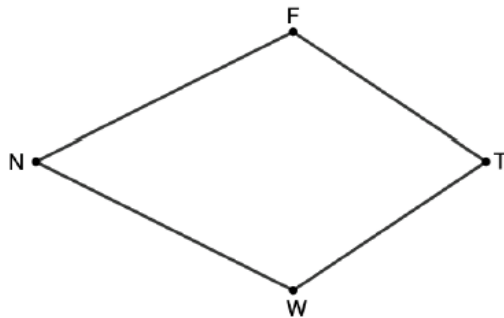
8.4.6 Wrap-Up: Cool Down

Component Overview: Students use the properties of kites to solve for an unknown variable.



8.4.6 Wrap-Up: Cool Down

- Kite $FTWN$ is shown. Find the value of x if $m\angle N = 53^\circ$, $m\angle T = 67^\circ$, and $m\angle F = \left(\frac{1}{2}x - 8\right)^\circ$.



$$m\angle N + m\angle T + m\angle F + m\angle W = 360^\circ$$

$$53^\circ + 67^\circ + \left(\frac{1}{2}x - 8\right)^\circ + \left(\frac{1}{2}x - 8\right)^\circ = 360^\circ$$

$$x + 104 = 360$$

$$x = 256$$



The information provided from the student here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.